Problem 2. Find a Möbius map that sends $0, 1, \infty$ to $1, i, -1$.

Solution. A short solution stands here. The figure of the next problem must not float above it.

Problem 3. Show that $F(z) = \dfrac{z-i}{z+i}$ maps the upper half-plane $\mathbb{H}$ conformally onto the unit disk $\mathbb{D}$.

Solution. For real x we have $|x-i| = |x+i|$, so F maps $\mathbb{R}$ into the unit circle; since $F(i) = 0 \in \mathbb{D}$ and $\mathbb{H}$ is connected, $F(\mathbb{H}) = \mathbb{D}$. In Figure 1, walking along $\mathbb{R}$ from A to C keeps $\mathbb{H}$ on the left, and its image keeps $\mathbb{D}$ on the left.

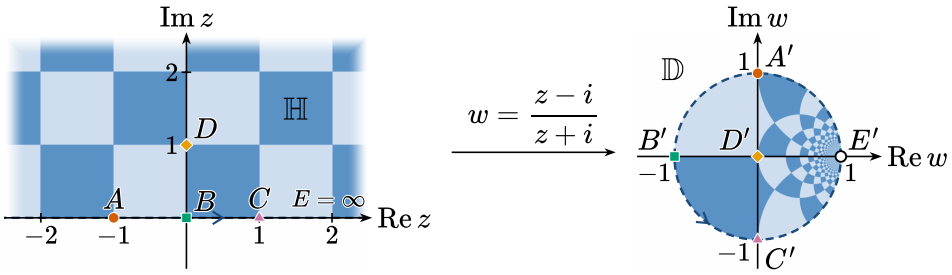


Figure 1: $w = (z-i)/(z+i)$ on $\mathbb{H}$: the points A–E and their images A'–E'. Dashed: boundary not included; arrowheads: orientation.

The figure stays here, right after the solution that cites it.